Integration

Lecture Notes for a Pure Mathematics Course

Calculus sequence: limits, continuity, differentiation, and integration

May 16, 2026

limits → continuity → differentiation → integration

Purpose of these notes

These notes present integration as a mathematical theory, not as a collection of recipes. The guiding questions are:

- How can area and accumulated change be defined rigorously?
- Why is integration the inverse process to differentiation?
- Which techniques allow us to compute integrals efficiently?
- What are the main mathematical applications of definite integrals?

The material assumes familiarity with limits, continuity, and differentiation.

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1 What integration is trying to solve

Differentiation begins with local behavior. Given a function F, its derivative F' measures the instantaneous rate of change of F. Integration begins with the reverse problem: from local values of a function, reconstruct a total quantity.

There are two central viewpoints:

1. **Inverse differentiation:** find functions F such that $F' = f$.
2. **Accumulation:** add infinitely many infinitesimal contributions $f(x)\,\mathrm{d}x$ over an interval.

The first viewpoint leads to indefinite integrals. The second leads to definite integrals. The Fundamental Theorem of Calculus explains why these two viewpoints are in fact two sides of the same theory.

1.1 A first geometric picture

If $f(x) \geq 0$ on $[a, b]$, then the definite integral

$$\int_a^b f(x)\,\mathrm{d}x$$

represents the area between the graph of f, the x-axis, and the vertical lines $x = a$, $x = b$.

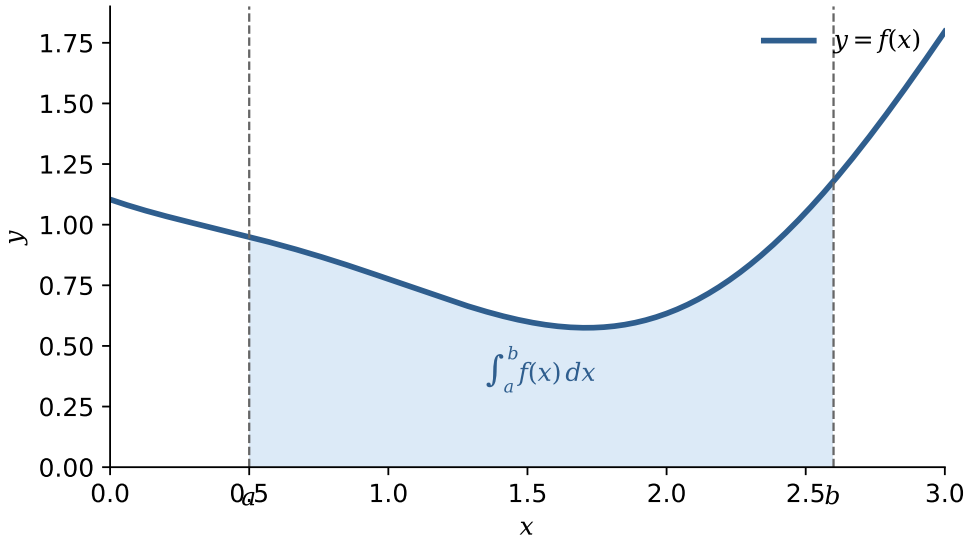

Figure 1: The definite integral as a signed area.

This picture is helpful, but it is not a definition. The definition will come from limits of sums.

1.2 Signed area

If f takes both positive and negative values, the integral counts area above the x-axis positively and area below the x-axis negatively. Thus the integral is not always ordinary geometric area; it is *signed area*.

For example,

$$\int_{-1}^{1} x\,\mathrm{d}x = 0,$$

even though the region between $y = x$ and the x-axis has positive geometric area. The positive part on $[0, 1]$ cancels the negative part on $[-1, 0]$.

Signed area versus total area

The integral $\int_a^b f(x)\,\mathrm{d}x$ gives signed area. Total geometric area is

$$\int_a^b |f(x)|\,\mathrm{d}x.$$

Confusing these two quantities is one of the most common errors in early integration.

2 Antiderivatives and indefinite integrals

2.1 Definition

Definition 2.1 (Antiderivative)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. A function $F : I \to \mathbb{R}$ is called an *antiderivative* of f on I if

$$F'(x) = f(x)$$

for every $x \in I$.

If F is one antiderivative of f, then so is $F + C$ for every constant $C \in \mathbb{R}$, because

$$(F + C)' = F'.$$

Proposition 2.2 (Antiderivatives differ by a constant)**.** *If $F'(x) = f(x)$ and $G'(x) = f(x)$ on an interval I, then there exists a constant C such that*

$$G(x) = F(x) + C$$

for all $x \in I$.

Proof. Let $H = G - F$. Then $H' = G' - F' = 0$ on I. By the Mean Value Theorem, a differentiable function with zero derivative on an interval is constant. Hence $H = C$, so $G = F + C$. □

2.2 Indefinite integral notation

The expression

$$\int f(x)\,\mathrm{d}x$$

denotes the set of all antiderivatives of f. In practice we write

$$\int f(x)\,\mathrm{d}x = F(x) + C,$$

where $F'(x) = f(x)$ and $C \in \mathbb{R}$.

The symbol $\mathrm{d}x$ is not decoration. It indicates the variable of integration. For instance,

$$\int 2x\,\mathrm{d}x = x^2 + C,$$

whereas

$$\int 2x\,\mathrm{d}t = 2xt + C$$

if x is treated as a constant and t is the integration variable.

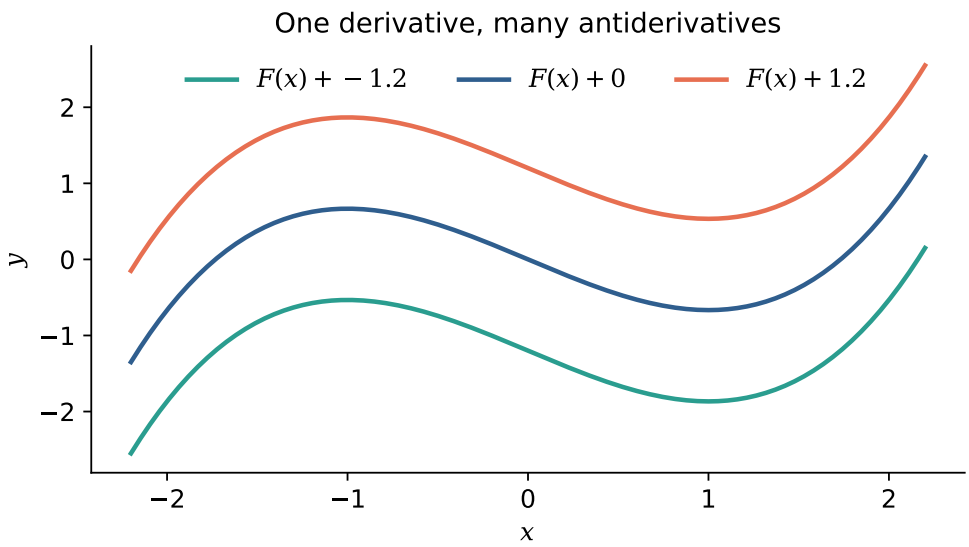

Figure 2: An antiderivative is not unique; constants shift the graph vertically.

2.3 Basic antiderivative table

The following table is derived from differentiation rules.

Function $f(x)$	Indefinite integral
x^n, $n \neq -1$	$\int x^n \, \mathrm{d}x = \frac{x^{n+1}}{n+1} + C$
$1/x$, $x \neq 0$	$\int \frac{1}{x} \, \mathrm{d}x = \ln \lvert x \rvert + C$
e^x	$\int e^x \, \mathrm{d}x = e^x + C$
a^x, $a > 0$, $a \neq 1$	$\int a^x \, \mathrm{d}x = \frac{a^x}{\ln a} + C$
$\cos x$	$\int \cos x \, \mathrm{d}x = \sin x + C$
$\sin x$	$\int \sin x \, \mathrm{d}x = -\cos x + C$
$\sec^2 x$	$\int \sec^2 x \, \mathrm{d}x = \tan x + C$
$1/(1+x^2)$	$\int \frac{1}{1+x^2} \, \mathrm{d}x = \arctan x + C$
$1/\sqrt{1-x^2}$	$\int \frac{1}{\sqrt{1-x^2}} \, \mathrm{d}x = \arcsin x + C$

2.4 Linearity

Indefinite integration is linear:

$$\int (af(x) + bg(x)) \, \mathrm{d}x = a \int f(x) \, \mathrm{d}x + b \int g(x) \, \mathrm{d}x,$$

for constants a, b.

Example 2.3. Compute

$$\int \left(4x^3 - 6x + 7\right) \, \mathrm{d}x.$$

Using linearity and the power rule,

$$\int \left(4x^3 - 6x + 7\right) \, \mathrm{d}x = x^4 - 3x^2 + 7x + C.$$

A useful check is differentiation:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^4 - 3x^2 + 7x + C\right) = 4x^3 - 6x + 7.$$

2.5 Initial conditions

An antiderivative becomes unique when one value is specified.

Example 2.4. Find F such that

$$F'(x) = 3x^2 - 4, \qquad F(1) = 5.$$

First find the general antiderivative:

$$F(x) = x^3 - 4x + C.$$

Use the condition:

$$F(1) = 1 - 4 + C = C - 3 = 5.$$

Thus $C = 8$, and

$$F(x) = x^3 - 4x + 8.$$

3 The definite integral as a limit of sums

3.1 Partitions

Let $[a, b]$ be a closed interval. A *partition* P of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_n = b.$$

The subintervals are

$$[x_{i-1}, x_i], \qquad i = 1, \ldots, n,$$

and their lengths are

$$\Delta x_i = x_i - x_{i-1}.$$

The mesh of the partition is

$$\|P\| = \max_{1 \le i \le n} \Delta x_i.$$

A tagged partition also chooses a point $x_i^* \in [x_{i-1}, x_i]$ in each subinterval.

3.2 Riemann sums

Given a bounded function $f : [a, b] \to \mathbb{R}$, a Riemann sum is

$$\sum_{i=1}^{n} f(x_i^*)\Delta x_i.$$

Each term has the form "height times width". The sum approximates the signed area under the graph.

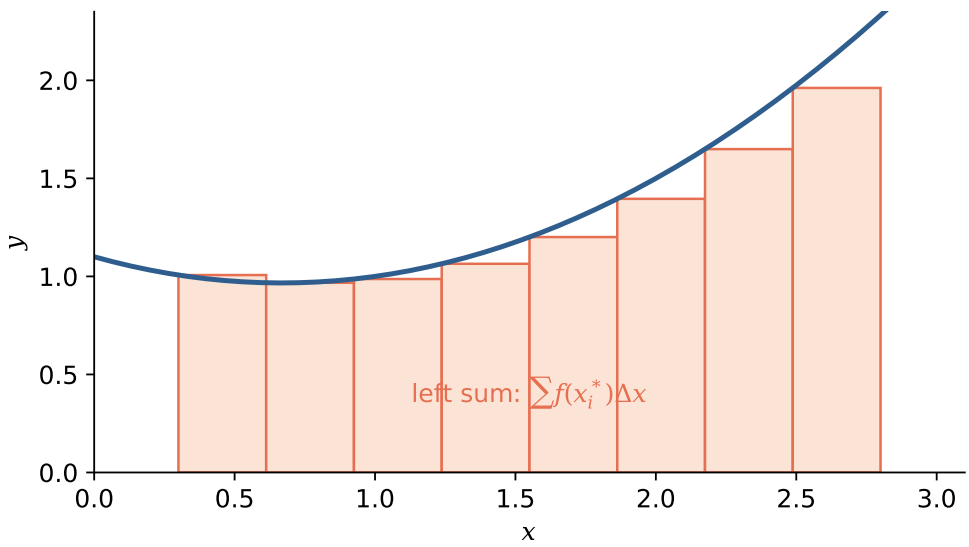

Figure 3: A Riemann sum approximates a curved area by rectangles.

3.3 Riemann integral

Definition 3.1 (Riemann integrability)**.** A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$ if there exists a number $I \in \mathbb{R}$ such that for every sequence of tagged partitions whose meshes tend to zero, the corresponding Riemann sums tend to I. We then write

$$\int_a^b f(x)\,\mathrm{d}x = I.$$

Intuitively, the value of the integral must be independent of how the tags are chosen, provided the partition becomes sufficiently fine.

3.4 Upper and lower sums

For a bounded function f, define on each subinterval

$$m_i = \inf_{x\in[x_{i-1},x_i]} f(x), \qquad M_i = \sup_{x\in[x_{i-1},x_i]} f(x).$$

The lower and upper sums are

$$L(f, P) = \sum_{i=1}^{n} m_i \Delta x_i, \qquad U(f, P) = \sum_{i=1}^{n} M_i \Delta x_i.$$

Upper-lower criterion

A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$, there exists a partition P such that

$$U(f, P) - L(f, P) < \varepsilon.$$

This criterion says that the best upper approximations and the best lower approximations can be made arbitrarily close.

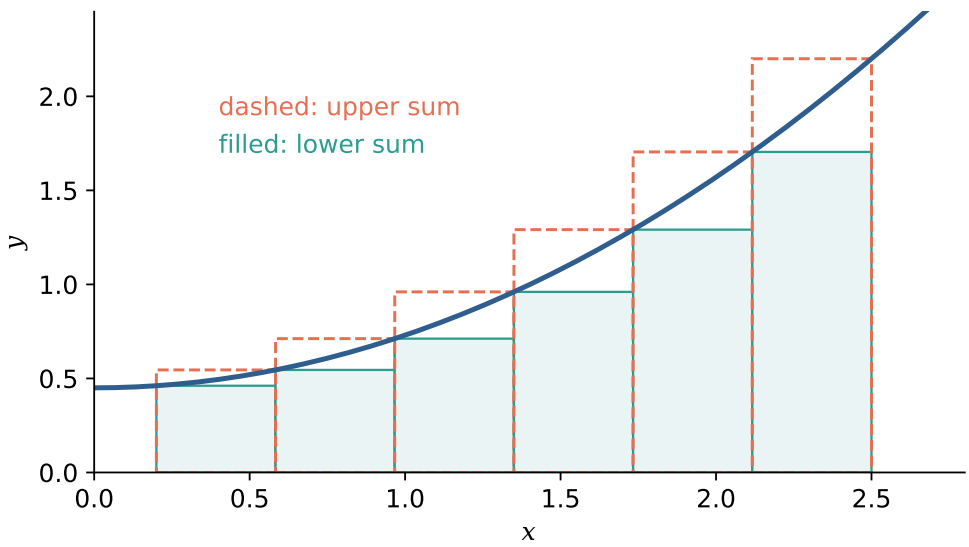

Figure 4: For a monotone function, lower and upper sums squeeze the true integral.

3.5 Important classes of integrable functions

The following facts are fundamental.

Theorem 3.2. *Every continuous function on a closed interval $[a, b]$ is Riemann integrable.*

Theorem 3.3. *Every monotone function on a closed interval $[a, b]$ is Riemann integrable.*

These results explain why most functions encountered in elementary calculus are integrable on closed intervals.

Integrability is not automatic

A bounded function may fail to be Riemann integrable. The standard example is the Dirichlet function

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

On every interval, its lower sum is 0 and its upper sum is 1 times the interval length. Hence upper and lower sums never meet.

4 Basic properties of definite integrals

Assume all functions involved are integrable on the relevant intervals.

4.1 Orientation and additivity

By convention,

$$\int_a^a f(x)\,\mathrm{d}x = 0$$

and

$$\int_b^a f(x)\,\mathrm{d}x = -\int_a^b f(x)\,\mathrm{d}x.$$

If $c \in [a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x.$$

4.2 Linearity

For constants α, β,

$$\int_a^b (\alpha f(x) + \beta g(x))\,\mathrm{d}x = \alpha \int_a^b f(x)\,\mathrm{d}x + \beta \int_a^b g(x)\,\mathrm{d}x.$$

4.3 Order properties

If $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x \leq \int_a^b g(x)\,\mathrm{d}x.$$

If $m \leq f(x) \leq M$ on $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x)\,\mathrm{d}x \leq M(b-a).$$

4.4 Absolute value inequality

If f is integrable, then

$$\left| \int_a^b f(x)\,\mathrm{d}x \right| \leq \int_a^b |f(x)|\,\mathrm{d}x.$$

This is the integral analogue of the triangle inequality.

4.5 Symmetry

If f is even, meaning $f(-x) = f(x)$, then

$$\int_{-a}^a f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x.$$

If f is odd, meaning $f(-x) = -f(x)$, then

$$\int_{-a}^a f(x)\,\mathrm{d}x = 0.$$

Example 4.1. Since $x^5 \cos x$ is odd, because x^5 is odd and $\cos x$ is even,

$$\int_{-2}^2 x^5 \cos x\,\mathrm{d}x = 0.$$

No antiderivative computation is needed.

5 The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus is the central bridge between differentiation and integration.

5.1 Accumulation functions

Given an integrable function f, define

$$A(x) = \int_a^x f(t)\,\mathrm{d}t.$$

The variable t is a dummy variable. The function A measures accumulated signed area from a to x.

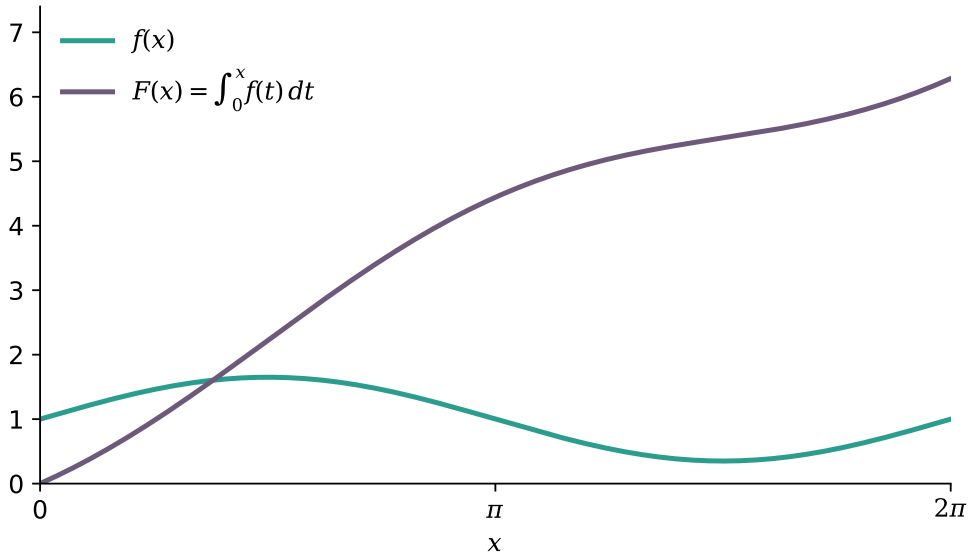

Figure 5: The accumulation function grows according to the values of f.

5.2 FTC, Part I

Theorem 5.1 (Fundamental Theorem of Calculus, Part I)**.** *Let f be continuous on an interval containing a. Define*

$$A(x) = \int_a^x f(t)\,\mathrm{d}t.$$

Then A is differentiable and

$$A'(x) = f(x).$$

Proof idea. Consider the difference quotient:

$$\frac{A(x+h) - A(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,\mathrm{d}t.$$

This is the average value of f on the small interval from x to $x+h$. Since f is continuous, as $h \to 0$ this average tends to $f(x)$. Hence $A'(x) = f(x)$. □

5.3 FTC, Part II

Theorem 5.2 (Fundamental Theorem of Calculus, Part II)**.** *Let f be continuous on $[a, b]$, and let F be any antiderivative of f on $[a, b]$. Then*

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

This is often written as

$$\int_a^b f(x)\,\mathrm{d}x = [F(x)]_a^b\,.$$

Example 5.3. Compute

$$\int_1^3 (2x+1)\,\mathrm{d}x.$$

An antiderivative is

$$F(x) = x^2 + x.$$

Therefore

$$\int_1^3 (2x+1)\,\mathrm{d}x = F(3) - F(1) = (9+3) - (1+1) = 10.$$

Two different constants

For indefinite integrals, always include C:

$$\int f(x)\,\mathrm{d}x = F(x) + C.$$

For definite integrals, the constant cancels:

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Do not write $+C$ at the end of a definite integral.

5.4 Differentiating an integral with variable limits

If

$$G(x) = \int_{\alpha(x)}^{\beta(x)} f(t)\,\mathrm{d}t,$$

where f is continuous and α, β are differentiable, then

$$G'(x) = f(\beta(x))\beta'(x) - f(\alpha(x))\alpha'(x).$$

Example 5.4. Let

$$G(x) = \int_{x^2}^{\sin x} e^{t^2}\,\mathrm{d}t.$$

Although e^{t^2} has no elementary antiderivative, the derivative of G is easy:

$$G'(x) = e^{\sin^2 x}\cos x - e^{x^4}\cdot 2x.$$

6 Average value and the mean value theorem for integrals

6.1 Average value of a function

The average value of an integrable function f on $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x.$$

This number is the constant height whose rectangle over $[a, b]$ has the same signed area as the integral of f.

6.2 Mean value theorem for integrals

Theorem 6.1. *If f is continuous on $[a, b]$, then there exists $c \in [a, b]$ such that*

$$\int_a^b f(x)\,\mathrm{d}x = f(c)(b-a).$$

Equivalently,

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x.$$

Proof. Since f is continuous on $[a, b]$, it attains a minimum m and a maximum M. Thus

$$m \leq \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x \leq M.$$

By the Intermediate Value Theorem, f assumes every value between m and M. Hence there exists $c \in [a, b]$ with the desired property. □

Example 6.2. Find the average value of $f(x) = x^2$ on $[0, 3]$.

$$f_{\text{avg}} = \frac{1}{3}\int_0^3 x^2\,\mathrm{d}x = \frac{1}{3}\left[\frac{x^3}{3}\right]_0^3 = \frac{1}{3}\cdot 9 = 3.$$

The theorem says there is $c \in [0, 3]$ such that $c^2 = 3$, namely $c = \sqrt{3}$.

7 Substitution

Substitution is the integral form of the chain rule.

7.1 The idea

The chain rule says

$$\frac{\mathrm{d}}{\mathrm{d}x}F(g(x)) = F'(g(x))g'(x).$$

If $F' = f$, then

$$\int f(g(x))g'(x)\,\mathrm{d}x = F(g(x)) + C.$$

Let $u = g(x)$. Then $\mathrm{d}u = g'(x)\,\mathrm{d}x$, so the integral becomes

$$\int f(u)\,\mathrm{d}u.$$

Substitution recipe

1. Identify an inner expression $u = g(x)$.
2. Compute $\mathrm{d}u = g'(x)\,\mathrm{d}x$.
3. Rewrite the entire integral in terms of u.
4. Integrate with respect to u.
5. For an indefinite integral, substitute back $u = g(x)$.

7.2 Indefinite substitution

Example 7.1. Compute

$$\int 2x\cos(x^2)\,\mathrm{d}x.$$

Let

$$u = x^2, \qquad \mathrm{d}u = 2x\,\mathrm{d}x.$$

Then

$$\int 2x\cos(x^2)\,\mathrm{d}x = \int \cos u\,\mathrm{d}u = \sin u + C = \sin(x^2) + C.$$

Example 7.2. Compute

$$\int \frac{x}{x^2+5}\,\mathrm{d}x.$$

Let $u = x^2 + 5$, so $\mathrm{d}u = 2x\,\mathrm{d}x$, hence $x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$. Therefore

$$\int \frac{x}{x^2+5}\,\mathrm{d}x = \frac{1}{2}\int \frac{1}{u}\,\mathrm{d}u = \frac{1}{2}\ln|u| + C = \frac{1}{2}\ln(x^2+5) + C.$$

7.3 Definite substitution

For a definite integral, one may either substitute back or change the limits.

Example 7.3. Compute

$$\int_0^1 6x(1+x^2)^2\,\mathrm{d}x.$$

Let $u = 1 + x^2$, so $\mathrm{d}u = 2x\,\mathrm{d}x$ and $6x\,\mathrm{d}x = 3\,\mathrm{d}u$. When $x = 0$, $u = 1$. When $x = 1$, $u = 2$. Thus

$$\int_0^1 6x(1+x^2)^2\,\mathrm{d}x = 3\int_1^2 u^2\,\mathrm{d}u = 3\left[\frac{u^3}{3}\right]_1^2 = 8 - 1 = 7.$$

A common substitution error

After setting $u = g(x)$, every occurrence of x and $\mathrm{d}x$ must be converted. A partially converted expression such as

$$\int \frac{x}{u}\,\mathrm{d}x$$

is not a valid final substitution.

8 Integration by parts

Integration by parts is the integral form of the product rule.

8.1 Derivation

The product rule says

$$(uv)' = u'v + uv'.$$

Integrating both sides gives

$$uv = \int u'v\,\mathrm{d}x + \int uv'\,\mathrm{d}x.$$

Rearranging,

$$\int uv' \, \mathrm{d}x = uv - \int u'v \, \mathrm{d}x.$$

Using differential notation $\mathrm{d}u = u' \, \mathrm{d}x$ and $\mathrm{d}v = v' \, \mathrm{d}x$, we obtain

$$\int u \, \mathrm{d}v = uv - \int v \, \mathrm{d}u.$$

Integration by parts

$$\boxed{\int u \, \mathrm{d}v = uv - \int v \, \mathrm{d}u}$$

For definite integrals,

$$\boxed{\int_a^b u \, \mathrm{d}v = [uv]_a^b - \int_a^b v \, \mathrm{d}u.}$$

8.2 Choosing u and $\mathrm{d}v$

A useful heuristic is to choose u as the factor that becomes simpler when differentiated, and choose $\mathrm{d}v$ as the remaining factor that can be integrated.

Example 8.1. Compute

$$\int xe^x \, \mathrm{d}x.$$

Choose

$$u = x, \qquad \mathrm{d}v = e^x \, \mathrm{d}x.$$

Then

$$\mathrm{d}u = \mathrm{d}x, \qquad v = e^x.$$

Thus

$$\int xe^x \, \mathrm{d}x = xe^x - \int e^x \, \mathrm{d}x = xe^x - e^x + C = e^x(x-1) + C.$$

Example 8.2. Compute

$$\int \ln x \, \mathrm{d}x, \qquad x > 0.$$

There is only one visible factor, so write $\ln x \, \mathrm{d}x = (\ln x) \cdot 1 \, \mathrm{d}x$. Choose

$$u = \ln x, \qquad \mathrm{d}v = \mathrm{d}x.$$

Then

$$\mathrm{d}u = \frac{1}{x} \, \mathrm{d}x, \qquad v = x.$$

Therefore

$$\int \ln x \, \mathrm{d}x = x \ln x - \int 1 \, \mathrm{d}x = x \ln x - x + C.$$

8.3 Repeated integration by parts

Example 8.3. Compute

$$\int x^2 e^x \, \mathrm{d}x.$$

First choose $u = x^2,\ \mathrm{d}v = e^x\,\mathrm{d}x$. Then

$$\int x^2 e^x\,\mathrm{d}x = x^2 e^x - \int 2xe^x\,\mathrm{d}x.$$

Apply integration by parts again to $\int 2xe^x\,\mathrm{d}x$:

$$\int 2xe^x\,\mathrm{d}x = 2\left(xe^x - \int e^x\,\mathrm{d}x\right) = 2xe^x - 2e^x.$$

Hence

$$\int x^2 e^x\,\mathrm{d}x = x^2 e^x - 2xe^x + 2e^x + C = e^x(x^2 - 2x + 2) + C.$$

9 Partial fractions

Partial fractions are used to integrate rational functions

$$R(x) = \frac{P(x)}{Q(x)},$$

where P, Q are polynomials.

9.1 Proper rational functions

A rational function is called *proper* if

$$\deg P < \deg Q.$$

If $\deg P \geq \deg Q$, first perform polynomial division.

Example 9.1. Decompose and integrate

$$\int \frac{5x+1}{x^2 - x - 2}\,\mathrm{d}x.$$

Factor the denominator:

$$x^2 - x - 2 = (x-2)(x+1).$$

Write

$$\frac{5x+1}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}.$$

Then

$$5x + 1 = A(x+1) + B(x-2).$$

Set $x = 2$: $11 = 3A$, so $A = 11/3$. Set $x = -1$: $-4 = -3B$, so $B = 4/3$. Therefore

$$\int \frac{5x+1}{x^2 - x - 2}\,\mathrm{d}x = \frac{11}{3}\ln|x-2| + \frac{4}{3}\ln|x+1| + C.$$

9.2 Repeated linear factors

If $(x-a)^k$ appears in the denominator, include all terms

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_k}{(x-a)^k}.$$

9.3 Irreducible quadratic factors

For an irreducible quadratic factor $ax^2 + bx + c$, use a numerator of degree one:

$$\frac{Ax + B}{ax^2 + bx + c}.$$

The resulting integrals often involve logarithms and arctangent functions.

Example 9.2. Compute

$$\int \frac{1}{x^2 + 4}\,\mathrm{d}x.$$

Since

$$x^2 + 4 = 4\left(1 + (x/2)^2\right),$$

let $u = x/2$, so $\mathrm{d}x = 2\,\mathrm{d}u$. Then

$$\int \frac{1}{x^2 + 4}\,\mathrm{d}x = \int \frac{2}{4(1 + u^2)}\,\mathrm{d}u = \frac{1}{2}\arctan u + C = \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C.$$

10 Trigonometric integrals

Trigonometric integrals are often simplified using identities.

10.1 Basic identities

The most important identities are

$$\sin^2 x + \cos^2 x = 1,$$

$$1 + \tan^2 x = \sec^2 x,$$

and the half-angle identities

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \qquad \cos^2 x = \frac{1 + \cos(2x)}{2}.$$

10.2 Odd powers

If a product contains an odd power of sine or cosine, save one factor and convert the rest using $\sin^2 x + \cos^2 x = 1$.

Example 10.1. Compute

$$\int \sin^3 x \cos^2 x\,\mathrm{d}x.$$

Write

$$\sin^3 x = \sin^2 x \sin x = (1 - \cos^2 x)\sin x.$$

Then

$$\int \sin^3 x \cos^2 x\,\mathrm{d}x = \int (1 - \cos^2 x)\cos^2 x \sin x\,\mathrm{d}x.$$

Let $u = \cos x$, $\mathrm{d}u = -\sin x\,\mathrm{d}x$. Therefore

$$\int \sin^3 x \cos^2 x\,\mathrm{d}x = -\int (1 - u^2)u^2\,\mathrm{d}u = -\int (u^2 - u^4)\,\mathrm{d}u.$$

So

$$\int \sin^3 x \cos^2 x\,\mathrm{d}x = -\frac{u^3}{3} + \frac{u^5}{5} + C = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C.$$

10.3 Even powers

Even powers often require half-angle identities.

Example 10.2. Compute

$$\int \cos^2 x \, \mathrm{d}x.$$

Using

$$\cos^2 x = \frac{1 + \cos(2x)}{2},$$

we obtain

$$\int \cos^2 x \, \mathrm{d}x = \frac{1}{2} \int 1 \, \mathrm{d}x + \frac{1}{2} \int \cos(2x) \, \mathrm{d}x = \frac{x}{2} + \frac{\sin(2x)}{4} + C.$$

11 Improper integrals

A definite integral is proper when the interval is finite and the integrand is bounded on that interval. If either condition fails, the integral is improper and must be defined by a limit.

11.1 Infinite intervals

Definition 11.1. If f is integrable on every interval $[a, R]$, define

$$\int_a^\infty f(x) \, \mathrm{d}x = \lim_{R \to \infty} \int_a^R f(x) \, \mathrm{d}x,$$

provided the limit exists as a finite number. In that case the improper integral converges.

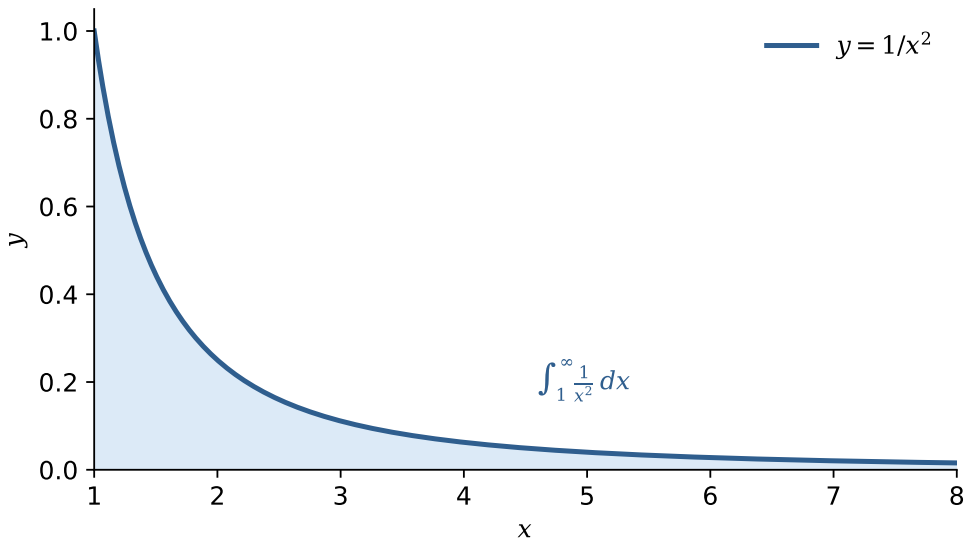

Figure 6: An improper integral over an infinite interval is defined by a limiting process.

Example 11.2. Determine whether

$$\int_1^\infty \frac{1}{x^2} \, \mathrm{d}x$$

converges. We compute

$$\int_1^R x^{-2} \, \mathrm{d}x = \left[-x^{-1}\right]_1^R = 1 - \frac{1}{R}.$$

Taking $R \to \infty$,

$$\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = 1.$$

The integral converges.

Example 11.3. The integral

$$\int_1^\infty \frac{1}{x}\,\mathrm{d}x$$

diverges, because

$$\int_1^R \frac{1}{x}\,\mathrm{d}x = \ln R \to \infty.$$

11.2 The p-integral test

The p-integral

For $p > 0$,

$$\int_1^\infty \frac{1}{x^p}\,\mathrm{d}x$$

converges if and only if $p > 1$. It diverges for $0 < p \leq 1$.

11.3 Vertical asymptotes

If f is unbounded near a, define

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{t \to a^+} \int_t^b f(x)\,\mathrm{d}x,$$

provided the limit exists.

Example 11.4. Determine whether

$$\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x$$

converges. We compute

$$\int_t^1 x^{-1/2}\,\mathrm{d}x = \left[2\sqrt{x}\right]_t^1 = 2 - 2\sqrt{t}.$$

Letting $t \to 0^+$, we obtain

$$\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = 2.$$

The integral converges despite the integrand being unbounded at 0.

Do not ignore improper behavior

The expression

$$\int_{-1}^1 \frac{1}{x}\,\mathrm{d}x$$

is not equal to 0 in the usual improper sense. It must be split at the singularity:

$$\int_{-1}^0 \frac{1}{x}\,\mathrm{d}x + \int_0^1 \frac{1}{x}\,\mathrm{d}x,$$

and both pieces diverge. Symmetric cancellation is not allowed unless one is explicitly using the Cauchy principal value.

12 Applications within calculus

12.1 Area between curves

If $g(x) \geq h(x)$ on $[a, b]$, then the area between the curves $y = g(x)$ and $y = h(x)$ is

$$A = \int_a^b (g(x) - h(x))\,\mathrm{d}x.$$

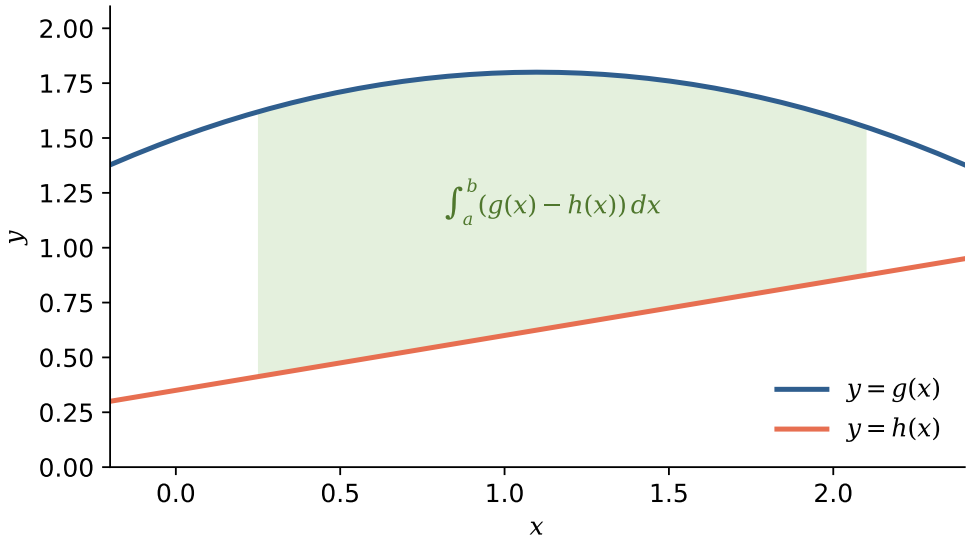

Figure 7: Area between curves is top minus bottom, integrated over the interval.

Example 12.1. Find the area between $y = 2x$ and $y = x^2$. First find the intersections:

$$x^2 = 2x \quad \Longleftrightarrow \quad x(x-2) = 0,$$

so $x = 0$ and $x = 2$. On $[0, 2]$, $2x \geq x^2$. Therefore

$$A = \int_0^2 (2x - x^2)\,\mathrm{d}x = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}.$$

12.2 Volumes by slicing

If a solid has cross-sectional area $A(x)$ perpendicular to the x-axis, then its volume is

$$V = \int_a^b A(x)\,\mathrm{d}x.$$

For solids of revolution, this leads to common formulas.

Disk method

If the region under $y = f(x) \geq 0$ from a to b is revolved around the x-axis, then

$$V = \pi \int_a^b f(x)^2\,\mathrm{d}x.$$

Washer method

If the outer radius is $R(x)$ and the inner radius is $r(x)$, then

$$V = \pi \int_a^b \left(R(x)^2 - r(x)^2 \right) \, \mathrm{d}x.$$

Example 12.2. The region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis. The volume is

$$V = \pi \int_0^4 (\sqrt{x})^2 \, \mathrm{d}x = \pi \int_0^4 x \, \mathrm{d}x = \pi \left[\frac{x^2}{2} \right]_0^4 = 8\pi.$$

12.3 Arc length

Let f be differentiable on $[a, b]$. Approximate the graph by short line segments. The length of each small segment is approximately

$$\sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x} \right)^2} \, \Delta x.$$

Passing to the limit gives

$$L = \int_a^b \sqrt{1 + (f'(x))^2} \, \mathrm{d}x.$$

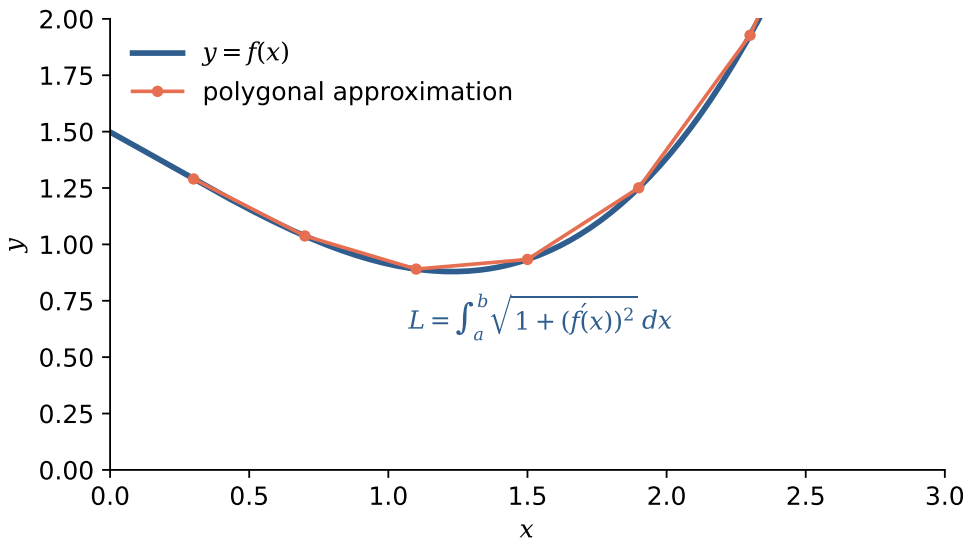

Figure 8: Arc length arises as the limit of polygonal approximations.

Example 12.3. Find the arc length of $f(x) = x^{3/2}$ on $[0, 1]$. First

$$f'(x) = \frac{3}{2}\sqrt{x}.$$

Thus

$$L = \int_0^1 \sqrt{1 + \frac{9}{4}x} \, \mathrm{d}x.$$

Let $u = 1 + \frac{9}{4}x$. Then $\mathrm{d}u = \frac{9}{4} \, \mathrm{d}x$, so $\mathrm{d}x = \frac{4}{9} \, \mathrm{d}u$. When $x = 0$, $u = 1$; when $x = 1$, $u = 13/4$. Hence

$$L = \frac{4}{9} \int_1^{13/4} u^{1/2} \, \mathrm{d}u = \frac{4}{9} \cdot \frac{2}{3} \left[u^{3/2} \right]_1^{13/4} = \frac{8}{27} \left(\left(\frac{13}{4} \right)^{3/2} - 1 \right).$$

13 A strategy for computing integrals

Not every integral has an elementary antiderivative. However, many course-level integrals are designed to be solved by recognizing a structure.

Practical decision tree

1. Simplify algebraically first. Expand, factor, split fractions, or use identities if appropriate.
2. Check whether the integrand is directly in the basic table.
3. Look for substitution: an inner function and its derivative nearby.
4. Look for integration by parts: product of functions where one simplifies after differentiation.
5. For rational functions, use division and partial fractions.
6. For powers of trigonometric functions, use identities and parity of powers.
7. For definite integrals, check symmetry before computing.
8. For improper integrals, identify the limiting process before evaluating.

13.1 Examples combining several ideas

Example 13.1. Compute

$$\int_0^1 x\sqrt{1+x^2}\,\mathrm{d}x.$$

Use substitution $u = 1 + x^2$, $\mathrm{d}u = 2x\,\mathrm{d}x$. Change limits: $u(0) = 1$, $u(1) = 2$. Then

$$\int_0^1 x\sqrt{1+x^2}\,\mathrm{d}x = \frac{1}{2}\int_1^2 u^{1/2}\,\mathrm{d}u = \frac{1}{2}\cdot\frac{2}{3}\left[u^{3/2}\right]_1^2 = \frac{1}{3}(2\sqrt{2}-1).$$

Example 13.2. Compute

$$\int_0^1 x\ln(1+x)\,\mathrm{d}x.$$

Use integration by parts:

$$u = \ln(1+x), \qquad \mathrm{d}v = x\,\mathrm{d}x.$$

Then

$$\mathrm{d}u = \frac{1}{1+x}\,\mathrm{d}x, \qquad v = \frac{x^2}{2}.$$

Thus

$$\int_0^1 x\ln(1+x)\,\mathrm{d}x = \left[\frac{x^2}{2}\ln(1+x)\right]_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{1+x}\,\mathrm{d}x.$$

Divide:

$$\frac{x^2}{1+x} = x - 1 + \frac{1}{1+x}.$$

Therefore

$$\int_0^1 x\ln(1+x)\,\mathrm{d}x = \frac{1}{2}\ln 2 - \frac{1}{2}\int_0^1 \left(x - 1 + \frac{1}{1+x}\right)\,\mathrm{d}x.$$

The remaining integral is

$$\left[\frac{x^2}{2} - x + \ln(1+x)\right]_0^1 = \frac{1}{2} - 1 + \ln 2 = \ln 2 - \frac{1}{2}.$$

Hence

$$\int_0^1 x\ln(1+x)\,\mathrm{d}x = \frac{1}{2}\ln 2 - \frac{1}{2}\left(\ln 2 - \frac{1}{2}\right) = \frac{1}{4}.$$

Example 13.3. Compute

$$\int_{-1}^{1} \frac{x^3}{1+x^2}\,\mathrm{d}x.$$

The function

$$f(x) = \frac{x^3}{1+x^2}$$

is odd, since the numerator is odd and the denominator is even. Therefore

$$\int_{-1}^{1} \frac{x^3}{1+x^2}\,\mathrm{d}x = 0.$$

14 Common mistakes and how to avoid them

Mistake 1: forgetting the constant

For indefinite integrals,

$$\int f(x)\,\mathrm{d}x = F(x) + C.$$

The constant is not optional. It represents the entire family of antiderivatives.

Mistake 2: changing limits incorrectly

If using substitution in a definite integral, either change the limits to the new variable or substitute back to the original variable before evaluating. Do not mix old limits with the new variable.

Mistake 3: treating improper integrals as proper

Before applying the Fundamental Theorem of Calculus, check whether the interval is finite and the function is bounded. If not, rewrite the integral as a limit.

Mistake 4: using area formulas without top-minus-bottom

For area between curves, identify which function is larger on each part of the interval. If the graphs cross, split the interval.

15 Worked problem set

Problem 1

Compute

$$\int (6x^5 - 4x^{-2} + 3e^x)\,\mathrm{d}x.$$

Solution.

$$\int (6x^5 - 4x^{-2} + 3e^x)\,\mathrm{d}x = x^6 + 4x^{-1} + 3e^x + C.$$

Problem 2

Compute

$$\int_0^2 (x^3 - 2x + 1)\,\mathrm{d}x.$$

Solution.

$$\int_0^2 (x^3 - 2x + 1)\,\mathrm{d}x = \left[\frac{x^4}{4} - x^2 + x\right]_0^2 = 4 - 4 + 2 = 2.$$

Problem 3

Compute

$$\int \frac{3x^2}{1 + x^3}\,\mathrm{d}x.$$

Solution. Let $u = 1 + x^3$. Then $\mathrm{d}u = 3x^2\,\mathrm{d}x$, so

$$\int \frac{3x^2}{1 + x^3}\,\mathrm{d}x = \int \frac{1}{u}\,\mathrm{d}u = \ln|u| + C = \ln\left|1 + x^3\right| + C.$$

Problem 4

Compute

$$\int x \sin x\,\mathrm{d}x.$$

Solution. Choose $u = x$, $\mathrm{d}v = \sin x\,\mathrm{d}x$. Then $\mathrm{d}u = \mathrm{d}x$ and $v = -\cos x$. Hence

$$\int x \sin x\,\mathrm{d}x = -x\cos x + \int \cos x\,\mathrm{d}x = -x\cos x + \sin x + C.$$

Problem 5

Find the area enclosed by $y = x$ and $y = x^2$.

Solution. The intersections are $x = x^2$, hence $x = 0$ and $x = 1$. On $[0, 1]$, $x \geq x^2$. Therefore

$$A = \int_0^1 (x - x^2)\,\mathrm{d}x = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Problem 6

Determine whether

$$\int_2^\infty \frac{1}{x \ln x}\,\mathrm{d}x$$

converges.

Solution. For $R > 2$,

$$\int_2^R \frac{1}{x \ln x}\,\mathrm{d}x = [\ln(\ln x)]_2^R = \ln(\ln R) - \ln(\ln 2).$$

As $R \to \infty$, $\ln(\ln R) \to \infty$. Thus the improper integral diverges.

16 Exercises for students

Basic antiderivatives

Compute the following indefinite integrals.

1. $\int (5x^4 - 3x^2 + 2)\,\mathrm{d}x$
2. $\int \left(\frac{4}{x} + 7e^x - 2\sin x\right)\,\mathrm{d}x$
3. $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)\,\mathrm{d}x$, for $x > 0$

Definite integrals

Compute.

1. $\int_0^1 (3x^2 + 2x)\,\mathrm{d}x$
2. $\int_{-2}^2 (x^4 - 3x^2 + 1)\,\mathrm{d}x$
3. $\int_0^\pi \sin x\,\mathrm{d}x$

Substitution

Compute.

1. $\int 4x(1 + 2x^2)^5\,\mathrm{d}x$
2. $\int \frac{\cos x}{1 + \sin x}\,\mathrm{d}x$
3. $\int_0^1 \frac{x}{1 + x^2}\,\mathrm{d}x$

Integration by parts

Compute.

1. $\int x\cos x\,\mathrm{d}x$
2. $\int x^2 \ln x\,\mathrm{d}x$, for $x > 0$
3. $\int_0^1 xe^{-x}\,\mathrm{d}x$

Improper integrals

Determine whether each integral converges, and evaluate it if it converges.

1. $\displaystyle\int_1^\infty \frac{1}{x^3}\,\mathrm{d}x$
2. $\displaystyle\int_0^1 \frac{1}{x^{2/3}}\,\mathrm{d}x$
3. $\displaystyle\int_0^1 \frac{1}{x}\,\mathrm{d}x$

17 Compact formula sheet

Core definitions and theorems

$$\int f(x)\,\mathrm{d}x = F(x) + C, \qquad F' = f,$$
$$\int_a^b f(x)\,\mathrm{d}x = \lim_{\|P\|\to 0} \sum_{i=1}^{n} f(x_i^*)\Delta x_i,$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(t)\,\mathrm{d}t = f(x),$$
$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Main techniques

$$\int f(g(x))g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u, \qquad u = g(x),$$
$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u,$$
$$\int_a^b (g(x) - h(x))\,\mathrm{d}x = \text{area between top } g \text{ and bottom } h,$$
$$\int_a^\infty f(x)\,\mathrm{d}x = \lim_{R\to\infty}\int_a^R f(x)\,\mathrm{d}x.$$

Integral	Result
$\int x^n\,\mathrm{d}x,\ n \neq -1$	$\frac{x^{n+1}}{n+1} + C$
$\int \frac{1}{x}\,\mathrm{d}x$	$\ln\lvert x\rvert + C$
$\int e^x\,\mathrm{d}x$	$e^x + C$
$\int \sin x\,\mathrm{d}x$	$-\cos x + C$
$\int \cos x\,\mathrm{d}x$	$\sin x + C$
$\int \sec^2 x\,\mathrm{d}x$	$\tan x + C$
$\int \frac{1}{1+x^2}\,\mathrm{d}x$	$\arctan x + C$